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Maternal and paternal smoking during pregnancy and the risk of ADHD in offspring: An umbrella review of systematic reviews and meta-analyses

Light Tsegay [a,1], Dawit Zena [b,1], Alemnesh Abicho [c], Yitbarek Gizachew [d], Getinet Ayano [e,*]

[a] Department of Medicine, Arsi University, Oromia, Ethiopia
[b] Department of Psychiatry, College of Health Sciences, Axum University, Ethiopia
[c] Ethiopian Public Health Institute, Addis Ababa, Ethiopia
[d] Bethel Medical College, Addis Ababa, Ethiopia
[e] School of Population Health, Curtin University, Australia

A B S T R A C T

Background: Prenatal exposure to maternal and paternal smoking has been investigated as a potential environmental risk factor for attention-deficit/hyperactivity disorder (ADHD) in offspring. Although many studies have examined these associations, the consistency and magnitude of the effect remain unclear. This umbrella review aimed to systematically evaluate and quantify the published evidence linking maternal and paternal smoking during pregnancy with ADHD risk in offspring.

Methods: We conducted an umbrella review of systematic reviews and meta-analyses. We systematically searched PubMed, Scopus, PsycINFO, and Web of Science from inception to September 21, 2024, and screened reference lists from relevant articles. We included systematic reviews and meta-analyses of observational studies that assessed the associations between maternal and paternal smoking during pregnancy and ADHD in offspring. Eligible meta-analyses used categorical ADHD diagnosis criteria based on DSM or ICD or less rigorous criteria such as self-reports. We excluded non-human studies, primary studies, genetic studies, and conference abstracts. We calculated summary odds ratios (ORs) with 95 % confidence intervals (CIs), heterogeneity using the I^2 statistic, 95 % prediction intervals, small study effects, and excess significance bias. Quality was assessed using the AMSTAR-2 tool. This study is registered with PROSPERO, number CRD42022361466.

Findings: From 2495 articles, six were eligible, yielding six meta-analyses that included a total of 2,218,965 participants for maternal smoking exposure and 183,666 participants for paternal smoking exposure. Maternal smoking was associated with a significantly increased risk of ADHD in offspring (OR 1.71, 95 % CI 1.46–2.00), with significant heterogeneity ($I^2 = 79$ %) and highly suggestive evidence (class II). Paternal smoking was also associated with a significantly increased ADHD risk (OR 1.36, 95 % CI 1.17–1.59), with moderate heterogeneity ($I^2 = 47$ %) and convincing evidence (class I). Further analyses indicated substantial heterogeneity and potential confounding by familial factors.

Conclusion: Both maternal and paternal smoking during pregnancy were significantly associated with an increased risk of ADHD in offspring. However, evidence from familial studies suggests that maternal smoking may be confounded by genetic and familial factors, underscoring the need for further high-quality research to establish causality. These findings highlight the importance of prenatal smoking prevention as a modifiable risk factor for reducing ADHD risk.

1. Introduction

The association between prenatal smoking, both maternal and paternal, and the risk of neurodevelopmental disorders, particularly attention-deficit/hyperactivity disorder (ADHD), has garnered significant attention in recent years (Ellis et al., 2012; Huang et al., 2018). ADHD is one of the most prevalent neurodevelopmental disorders worldwide, affecting approximately 8 % of children, and its causes are believed to be multifactorial, involving both genetic and environmental risk factors (Ayano et al., 2021, 2023; Cortese et al., 2012). Prenatal smoking is a modifiable environmental risk factor that has been extensively studied due to its harmful effects on fetal development, including the potential to disrupt normal brain maturation (Knopik et al., 2012; Gilman et al., 2008). Understanding the impact of prenatal smoking on ADHD risk is critical, as it provides opportunities for public health interventions to reduce preventable cases of ADHD.

Maternal smoking during pregnancy has been linked to a variety of adverse health outcomes in offspring, including low birth weight, preterm birth, and an increased risk of respiratory and cognitive problems (Gilman et al., 2008; Pereira et al., 2017; McEvoy and Spindel, 2017). In the context of ADHD, maternal smoking is hypothesized to interfere with fetal brain development through multiple mechanisms, including nicotine exposure, which affects the cholinergic system; hypoxia due to reduced oxygen supply from carbon monoxide in cigarette smoke; and

* Corresponding author. School of Population Health, Curtin University, WA, Australia.
E-mail addresses: babiget2015@gmail.com, g.yaya@curtin.edu.au (G. Ayano).
[1] Joint first authors.

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immune activation, which can result in inflammation that crosses the placenta and potentially alters fetal neurodevelopment (Bouwland et al., 2015; Suter et al., 2012). Despite these biological hypotheses, the strength of the association between maternal smoking and ADHD remains debated, with some studies suggesting a robust link, while others report that the association diminishes after adjusting for genetic and familial factors (Thapar et al., 2009; Joelsson et al., 2016).

In contrast, paternal smoking has received less attention in the literature, although it may also contribute to the risk of ADHD through both direct and indirect pathways. Direct effects could include second-hand smoke exposure to the mother and fetus, while indirect effects may involve epigenetic changes in sperm caused by smoking, leading to altered gene expression in the offspring (Langley et al., 2012; Joehanes et al., 2016). Moreover, the shared genetic predisposition between fathers who smoke and their children could confound the relationship, making it unclear whether the observed associations are due to smoking itself or underlying familial traits (Gustavson et al., 2017).

Although numerous cohort and case-control studies have investigated the relationship between prenatal smoking and ADHD, the evidence is fragmented across various populations, study designs, and outcomes. Systematic reviews (SRs) and meta-analyses have been conducted to synthesise these findings, yet they often present conflicting conclusions (Thapar et al., 2009; Joelsson et al., 2016; He et al., 2020). Additionally, the methodological quality of these reviews, including the handling of confounding factors, the rigor of statistical methods, and the assessment of heterogeneity, has not been consistently evaluated. This lack of a comprehensive synthesis makes it challenging to draw definitive conclusions about the true strength and credibility of the association between prenatal smoking and ADHD risk.

Against this background, an umbrella review offers an opportunity to systematically evaluate the existing SRs and meta-analyses to determine the overall strength of evidence on this topic. Umbrella reviews, which are overviews of SRs and meta-analyses, provide a higher-level synthesis that can help to identify whether the evidence base is consistent or contradictory, and whether the conclusions drawn from individual reviews are reliable. This approach allows for a more robust assessment of the quality of the evidence and the strength of the associations reported across studies.

Therefore, the aim of this umbrella review is to provide a comprehensive overview of the breadth and validity of the associations between maternal and paternal smoking during pregnancy and attention-deficit/hyperactivity disorder (ADHD) in offspring. Specifically, we seek to synthesise the magnitude, consistency, and credibility of these associations across the existing evidence base, including an assessment of heterogeneity and potential biases using established umbrella review criteria (Ayano et al., 2024; Belbasis et al., 2015; Ioannidis and Trikalinos, 2007). By doing so, this umbrella review aims to clarify whether prenatal smoking should be considered an important, potentially modifiable risk factor for ADHD and to identify priority targets for future clinical, preventive, and policy interventions.

2. Methods

2.1. Study design and eligibility criteria

We employed an umbrella review methodology to systematically identify, appraise, and synthesise all available systematic reviews (SRs) and meta-analyses on this topic. An umbrella review is a systematic overview of SRs and meta-analyses that provides a high-level synthesis of a broad and heterogeneous evidence base and enables formal grading of the strength and consistency of associations across studies. (Aromataris et al., 2015; Slim and Marquillier, 2022). The review protocol was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO; CRD42022361466).

Eligible SRs were those that assessed the association between prenatal smoking exposure (maternal or paternal) and ADHD outcomes, and that conducted a quantitative meta-analysis of cohort or case-control studies. Included reviews were required to report study-level effect sizes (ESs) with corresponding 95 % confidence intervals (CIs). SRs that did not include a meta-analysis, lacked extractable ESs, or synthesised evidence narratively only were excluded. When multiple SRs covered overlapping primary studies, the review with the largest and most comprehensive dataset was retained, consistent with recommended umbrella review methods (Ioannidis, 2009).

2.2. Literature search, systematic review selection, and data extraction

We systematically searched the following databases from inception to September 21, 2024: Medline, Scopus, PsychINFO, and Web of Science. The search strategy was designed to capture SRs and meta-analyses related to maternal and paternal smoking exposure and ADHD in offspring, with no language restrictions. The search was supplemented by manual screening of reference lists from included studies. The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for the selection of SRs (Moher et al., 2009).

Two reviewers (GA and DZ) independently screened titles and abstracts for eligibility, with disagreements resolved by a third reviewer through discussion. Full-text reviews were conducted for articles that could not be definitively included or excluded based on title and abstract review. From the included SRs, two investigators independently extracted data, including first author, year of publication, number of included studies, outcomes, and summary meta-analytic estimates.

2.3. Quality assessment of included systematic reviews

The methodological quality of the included SRs was assessed using the AMSTAR-2 tool (A Measurement Tool to Assess Systematic Reviews), a 16-point checklist used to evaluate the rigor of SRs (Shea et al., 2017). Two independent reviewers applied the AMSTAR-2 tool to each SR, with discrepancies resolved by discussion. The assessment included key criteria such as protocol registration, comprehensive search strategies, and appropriate methods for assessing risk of bias (Shea et al., 2017).

2.4. Statistical analysis

Summary odds ratios (ORs) with 95 % CIs were re-estimated using a common metric and random-effects models, anticipating high heterogeneity among studies (DerSimonian and Laird, 1986). Heterogeneity was quantified using Cochran's Q test and the I^2 statistic, with p-values <0.10 considered statistically significant (Higgins et al., 2003). The 95 % prediction interval (PI) was calculated to account for between-study heterogeneity and estimate the range of effects expected in a new study (IntHout et al., 2016). Egger's test was applied to evaluate publication bias and small-study effects, with p-values ≤ 0.10 indicating potential bias (Egger et al., 1997).

We also tested for excess significance bias, comparing the number of studies with statistically significant results (O) to the expected number (E), based on power calculations for each study (Ioannidis and Trikalinos, 2007). Associations were classified according to strength of evidence as "convincing," "highly suggestive," "suggestive," or "weak," based on statistical criteria, including heterogeneity, sample size, and significance (Schmid et al., 2020). Statistical analyses were performed using Stata version 12.0 (Stata Corp, College Station, TX, USA), and all p-values were two-tailed (Schmid et al., 2020; Furukawa and Leucht, 2011).

3. Results

From the six eligible meta-analyses identified in the umbrella review (out of 2495 articles initially identified), the total data included 2,218,965 participants for maternal smoking exposure and 183,666

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participants for paternal smoking exposure (Fig. 1) (Huang et al., 2018; He et al., 2020; Dong et al., 2018; Godleski et al., 2024; Haan et al., 2022; Maher et al., 2024). These meta-analyses focused on ADHD as the outcome, with a total of 172,987 ADHD cases (159,765 cases for maternal smoking and 13,222 cases for paternal smoking). The median number of ADHD cases per meta-analysis was 16,000 (IQR 2500 to 85,000), and the median total population per meta-analysis was 198,000 (IQR 50,000 to 925,000).

Four of the six meta-analyses were based on cohort studies, while two included a combination of cohort, case-control, or cross-sectional studies. The median number of study estimates per meta-analysis was 7 (IQR 5 to 9, range 3–15). Both associations, for maternal and paternal smoking, were statistically significant under the random-effects model. For maternal smoking, the random effects summary estimate was OR 1.71 (95 % CI: 1.46–2.00), with a p-value of 3·8E-21. The heterogeneity was high, with $I^2 = 79$ %, and the 95 % prediction interval ranged from 1.05 to 2.75. For paternal smoking, the random effects summary estimate was OR 1.36 (95 % CI: 1.17–1.59), with a p-value of 1·6E-29. The heterogeneity was moderate, with $I^2 = 47$ %, and the 95 % prediction interval ranged from 1.08 to 2.24 (Table 1).

The Egger test for publication bias showed a significant effect for maternal smoking ($p = 0.004$) but was non-significant for paternal smoking ($p = 0.42$). Both associations were robust under a 10 % credibility ceiling, with maternal smoking showing high heterogeneity and paternal smoking showing moderate heterogeneity. The level of evidence for maternal smoking was classified as Highly Suggestive (Class II), while paternal smoking was classified as Convincing (Class I) (Table 1).

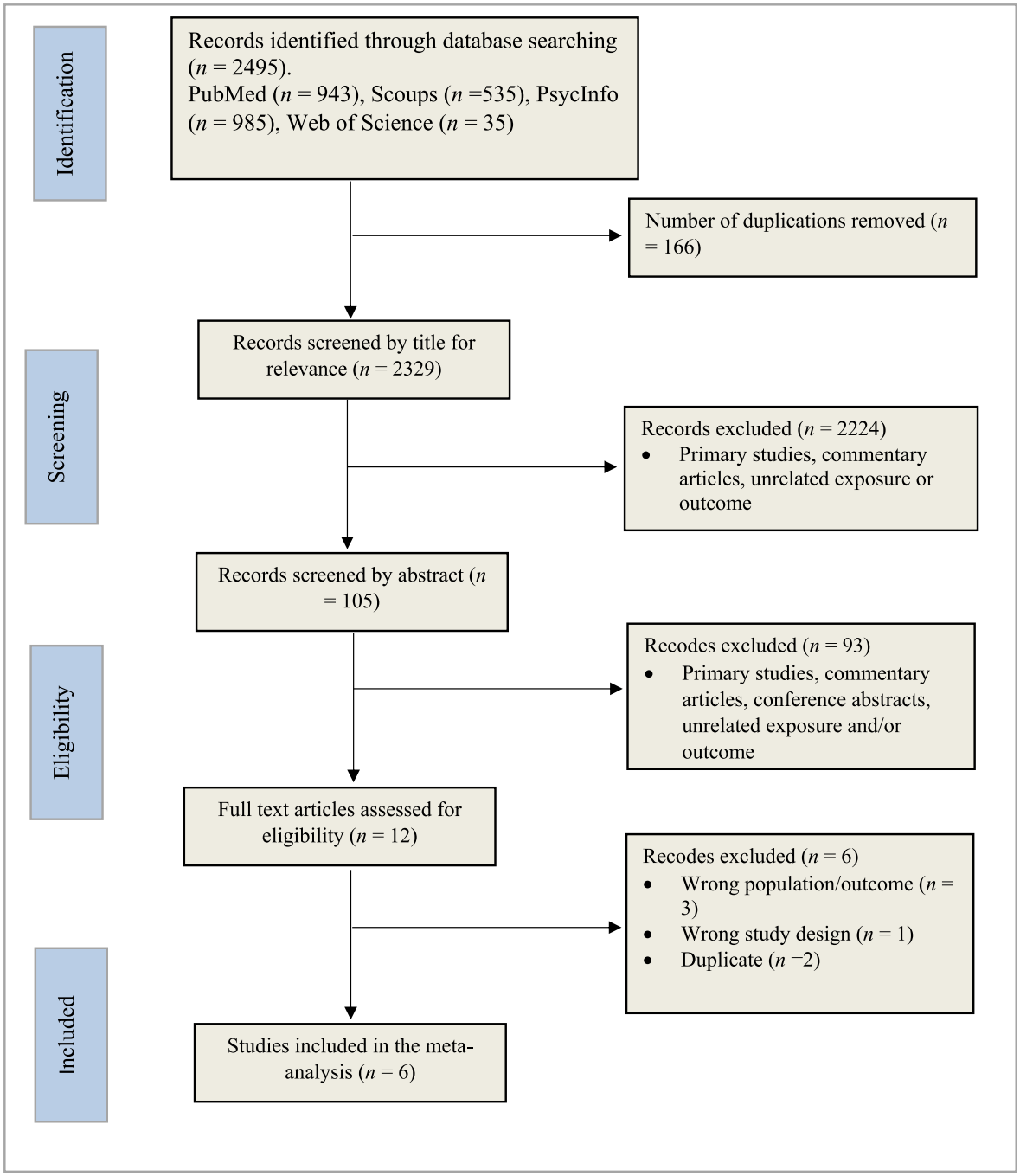


Fig. 1. PRISMA flowchart of included studies.

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Table 1
Maternal and Paternal Prenatal Smoking and ADHD in Offspring.

Exposure	Author, Year	Number of Cases/Total Population	Number of Study Estimates	Study Design	Effect Metric	Random Effects Summary Estimate (95 % CI)	Random Effects P-value	I (Huang et al., 2018)	95 % Prediction Interval	Egger P-value	Heterogeneity, Study Effects, etc.	AMSTAR 2 Quality	Level of Evidence
Maternal Smoking During Pregnancy	Dong et al., 2018	85,684/ 1,184,497	16	Cohort, case-control	OR	1.71 (1.46–2.00)	3·8E-21	79 %	1.05 to 2.75	0.004	Large heterogeneity *	High/High	Highly Suggestive (Class II)
Paternal Smoking During Pregnancy	Haan et al., 2021	13,222/ 183,666	4	Cohort, case-control	OR	1.36 (1.17–1.59)	1·6E-29	47 %	1.08 to 2.24	0.42	Moderate heterogeneity	High/High	Convincing (Class I)

Note. Table 1 summarises the results of umbrella-level meta-analyses evaluating maternal and paternal prenatal smoking in relation to ADHD in offspring. For each exposure, the table reports the total number of cases and participants, number of study estimates, study designs, random-effects pooled odds ratios (ORs) with 95 % confidence intervals (CIs), heterogeneity (I^2 and 95 % prediction interval), small-study effects (Egger's test), qualitative assessment of heterogeneity, AMSTAR 2 quality rating, and the corresponding level of evidence based on pre-specified umbrella review criteria. ADHD = attention-deficit/hyperactivity disorder; OR = odds ratio; CI = confidence interval.

3.1. Quality of the studies

The quality of the six meta-analyses included in this umbrella review was assessed using the AMSTAR 2 tool. Three of the studies were classified as high quality, as they met most of the AMSTAR 2 criteria, including having clear research questions, comprehensive literature search strategies, and appropriate methods for study selection and data extraction (Appendix 2). These studies also conducted thorough risk of bias (RoB) assessments, reported heterogeneity, and performed adequate analyses of publication bias, such as small study effects. Two studies were categorized as moderate quality, as they lacked important elements such as protocol registration, detailed exclusion criteria, and, in some cases, assessments of RoB and publication bias. Additionally, some deviations from the protocol were noted without sufficient justification, which impacted their overall quality rating. One study was rated as low quality, as it failed to meet several key AMSTAR 2 criteria, including duplicate study selection, data extraction, risk of bias assessment, and heterogeneity analysis. Furthermore, this study lacked transparency in reporting funding sources and potential conflicts of interest. Overall, 50 % of the meta-analyses were rated as high quality, while 33 % were moderate, and 17 % were low quality (Appendix 2). These findings reflect a range of methodological rigor across the included reviews, from robust to weaker standards.

3.2. Level of evidence for the associations

The hierarchical grading of evidence from the six meta-analyses on maternal and paternal smoking exposure during pregnancy in relation to ADHD in offspring was based on statistical significance, heterogeneity, and robustness of the associations. Paternal smoking during pregnancy was graded as convincing evidence (Class I). This association had a random effects summary estimate of OR 1.36 (95 % CI: 1.17–1.59), with a p-value of 1·6E-29, moderate heterogeneity ($I^2 = 47$ %), and a 95 % prediction interval of 1.08–2.24, which did not cross the null. The strength of evidence was supported by its large sample size, absence of small study effects, and the exclusion of excess significance bias.

Maternal smoking during pregnancy was graded as highly suggestive evidence (Class II). The random effects summary estimate was OR 1.71 (95 % CI: 1.46–2.00), with a p-value of 3·8E-21, but the heterogeneity was high ($I^2 = 79$ %). The 95 % prediction interval ranged from 1.05 to 2.75, indicating that the association remained statistically significant. While maternal smoking showed large heterogeneity, it retained significance under a 10 % credibility ceiling, but small study effects and excess significance bias were present, which contributed to the classification as Class II evidence (Table 1).

3.3. Sensitivity analysis by study design and adjustment for psychiatric history

The sensitivity analysis, stratified by both study design and adjustment for psychiatric history, showed consistent associations between both maternal and paternal smoking exposure and the risk of ADHD in offspring. For maternal smoking, the pooled odds ratio (OR) from cohort studies was 1.84 (95 % CI: 1.34–2.53) based on 25 studies, while case-control studies showed a pooled OR of 1.92 (95 % CI: 1.47–2.44) from 18 studies. Similarly, for paternal smoking, cohort studies had a pooled OR of 1.36 (95 % CI: 1.15–1.61) from 4 studies, while case-control studies yielded a higher pooled OR of 1.88 (95 % CI: 1.24–2.81) from 2 studies.

Further stratification by adjustment for psychiatric history revealed higher effect estimates for unadjusted analyses, suggesting the presence of potential confounding effects. For maternal smoking, the pooled OR for adjusted studies was 1.76 (95 % CI: 1.35–2.30) across 22 studies, while unadjusted studies yielded a higher pooled OR of 1.92 (95 % CI: 1.55–2.40) from 21 studies. For paternal smoking, the pooled OR for adjusted studies was 1.34 (95 % CI: 1.15–1.56) from 3 studies, while

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unadjusted studies showed a higher pooled OR of 1.88 (95 % CI: 1.25–2.81) from 3 studies.

4. Discussion

This umbrella review systematically and quantitatively synthesised the hierarchy of evidence for maternal and paternal smoking as risk factors for ADHD in offspring. Overall, prenatal smoking was associated with an increased risk of ADHD, but the strength and consistency of evidence differed by parental source. Paternal smoking during pregnancy showed convincing evidence (Class I) of association with offspring ADHD (OR 1.36, 95 % CI 1.17–1.59; $I^2 = 47$ %), whereas maternal smoking was supported by highly suggestive evidence (Class II; OR 1.71, 95 % CI 1.46–2.00) with substantial heterogeneity ($I^2 = 79$ %). Taken together, these findings indicate that prenatal smoking is a robust marker of ADHD risk in offspring, but they also highlight important uncertainty regarding causality and the specific contribution of maternal versus paternal pathways.

The convincing evidence for paternal smoking suggests a moderate but reliable increase in ADHD risk. Several mechanisms may underlie this association, including epigenetic alterations in sperm (e.g. DNA methylation changes affecting neurodevelopmental genes), oxidative stress–related sperm DNA damage, and increased second-hand smoke exposure of pregnant partners and fetuses (Agarwal et al., 2014; Jaakkola and Gissler, 2004) (Ernst et al., 2001). In addition, the association may partly reflect shared familial liability: fathers who smoke are more likely to share genetic and environmental risk factors with children who develop ADHD, including impulsivity, externalising traits, or broader familial psychopathology (Thapar et al., 2009). Thus, paternal smoking may act both as a potential biological exposure and as a proxy marker for intergenerational genetic and environmental risk.

By contrast, maternal smoking during pregnancy showed a stronger point estimate but marked between-study variability. Biologically, maternal smoking is highly plausible as a risk factor: nicotine and other tobacco constituents can disrupt fetal brain development through effects on the cholinergic system, fetal hypoxia, epigenetic modifications, and immune activation, all of which may interfere with neural circuits underlying attention, impulse control, and emotion regulation (Lewis et al., 2022; Lopes, 2019; Dietert and Zelikoff, 2008; Sussman, 2002; Ekblad et al., 2015). However, the substantial heterogeneity ($I^2 = 79$ %) suggests that the size—and possibly the presence—of the association varies meaningfully across contexts and study designs. This variability is likely driven by multiple factors, including differences in how smoking exposure (timing, intensity, cessation) and ADHD outcomes (diagnostic criteria, informants, age at assessment) are measured; variation in adjustment for key confounders such as maternal ADHD, depression, socioeconomic status, and co-occurring substance use; and differences in underlying population risk profiles. Importantly, studies using more rigorous causal designs (e.g. sibling comparisons or negative control analyses) often report attenuated associations, implying that part of the observed link between maternal smoking and ADHD may be explained by shared familial or genetic factors rather than a purely intrauterine causal effect.

These nuances have important conceptual and practical implications. Conceptually, our findings support the view that prenatal smoking is not simply an isolated teratogenic exposure but part of a broader intergenerational risk context for ADHD. The stronger and more consistent evidence for paternal smoking, together with heterogeneous evidence for maternal smoking, points towards a combination of genetic liability, shared environmental factors (e.g. family stress, socioeconomic disadvantage), and direct toxic effects of tobacco smoke. Clinically and from a public health perspective, this underscores the importance of addressing smoking as a *family-level* risk factor, rather than focusing solely on pregnant women. Integrating smoking cessation support for both parents into preconception and antenatal care may therefore be a pragmatic strategy to reduce a cluster of adverse outcomes, including ADHD, even if the causal fraction attributable to intrauterine exposure remains uncertain.

In practice, these findings reinforce existing recommendations for universal smoking cessation during the perinatal period, but add nuance by emphasising paternal involvement. For clinicians, prenatal smoking—maternal or paternal—may serve as a useful marker of elevated developmental risk, prompting closer monitoring of child behaviour, early identification of attentional and behavioural difficulties, and timely referral to support services. At the policy level, our results support continued investment in tobacco control measures and targeted cessation programs in reproductive-age populations, particularly in families with known ADHD or other psychiatric vulnerabilities.

This umbrella review also highlights key gaps and priorities for future research. First, there is a need for more high-quality, causally informative designs—such as sibling-comparison studies, negative control parent designs, adoption studies, and Mendelian randomisation—to better disentangle intrauterine effects from shared familial liability.45,49 Second, future meta-analyses should systematically stratify by key methodological features (e.g. degree of confounder adjustment, exposure timing and intensity, ADHD assessment method) to clarify sources of heterogeneity in maternal smoking results and identify subgroups or contexts where the association is strongest. Third, mechanistic studies integrating epigenetic, neuroimaging, and longitudinal phenotypic data could help link prenatal smoking exposures to specific neurodevelopmental pathways implicated in ADHD. Finally, more work is needed in diverse populations and low- and middle-income settings, where patterns of smoking, co-exposures, and healthcare access may differ substantially, to ensure that inferences and prevention strategies are globally relevant.

This umbrella review has limitations inherent to synthesising observational evidence, including the possibility of residual confounding, exposure misclassification, and publication or small-study bias.48–50 Our evidence grading did not fully account for biological plausibility or explicitly reward quasi-experimental designs, which may lead to conservative classification of some associations. Nonetheless, by applying a structured hierarchy of evidence across multiple SRs and meta-analyses, this review provides a comprehensive and nuanced assessment of the links between maternal and paternal smoking and offspring ADHD. The findings highlight prenatal smoking as an important intergenerational risk marker for ADHD and underscore the need for both more rigorous causal research and family-focused smoking cessation strategies during the preconception and antenatal periods (Krieger et al., 2010).

5. Conclusion

In conclusion, both maternal and paternal smoking during pregnancy were identified as significant risk factors for ADHD in offspring, with paternal smoking showing convincing evidence (Class I) and maternal smoking showing highly suggestive evidence (Class II). These findings, while robust, underscore the need for cautious interpretation and further research, particularly with regard to potential confounding factors and the mechanisms underlying these associations.

CRediT authorship contribution statement

Light Tsegay: Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Dawit Zena:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Alemnesh Abicho:** Writing – review & editing, Visualization, Supervision. **Yitbarek Gizachew:** Writing – review & editing, Writing – original draft, Validation, Data curation. **Getinet Ayano:** Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

None.

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Appendix A. Supplementary data

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